

Library System

AI-Integrated University Library Platform

Üsküdar University · Full-Stack · Node.js Monorepo

NestJS · Next.js 14 · PostgreSQL · Prisma · OpenRouter AI · Docker

What Is the Library System?

A full-stack Node.js monorepo built for Üsküdar University, covering every aspect of modern library management — from catalog browsing and physical borrowing to AI-powered study assistance.

apps/api

NestJS 10 · Prisma ORM · PostgreSQL · JWT Auth · Swagger · S3 · AI Services

apps/web

Next.js 14 App Router · React 18 · Tailwind CSS · SWR · Framer Motion

4

User Roles

20

Backend Modules

30+

Frontend Routes

100+

API Endpoints

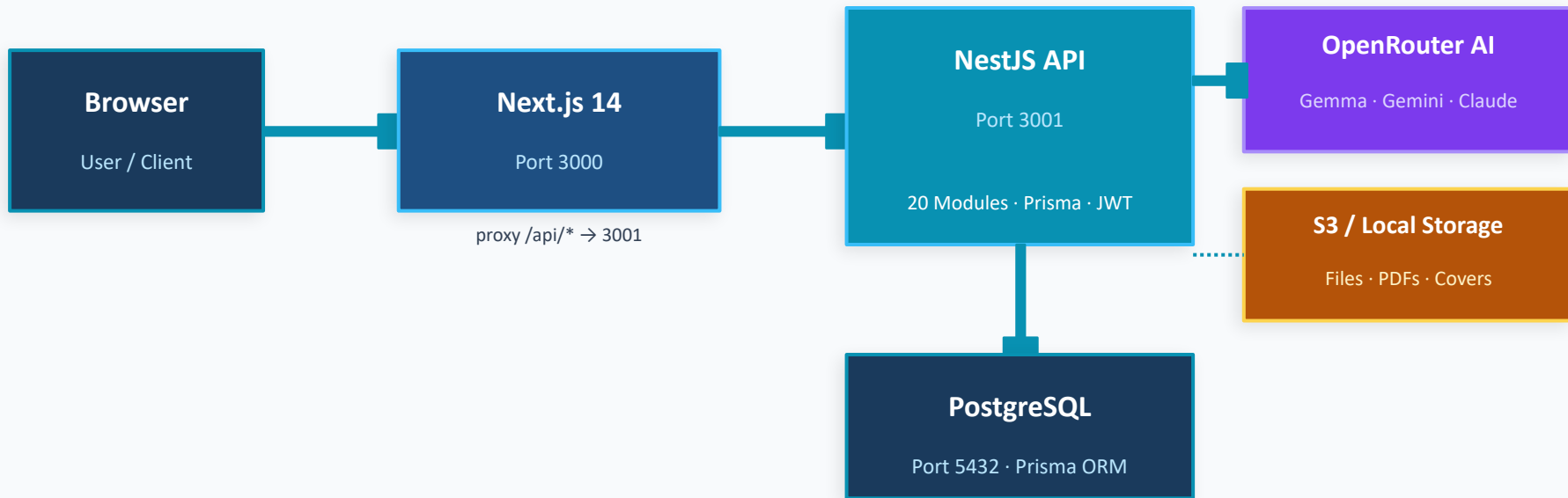
15

AI Tools

14

Test Suites

System Architecture



Frontend

Backend

AI Provider

Role System & Auth

STUDENT

Catalog · Borrow · Reserve
Fines · AI Assistant
Reading Lists · Notifications

INSTRUCTOR

All Student features
Upload Materials for
Approval
Create Reading Lists ·
Followers

STAFF

Catalog · Borrows
Reservations · AI Assistant
Profile & Notifications

ADMIN

Full system access
Users · Policies · Reports
Branches · AI Cover Scan

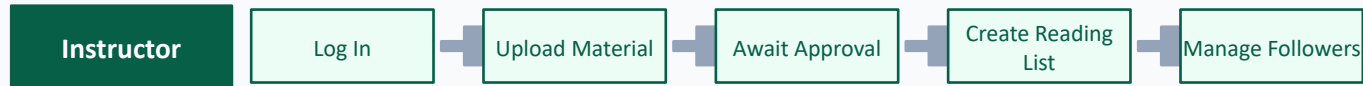
Authentication:

Email/Password + Email Verification · Google OAuth (optional) · HttpOnly JWT Cookie · Role-gated middleware (frontend + backend guards)

User Journeys by Role



+ AI Assistant is available throughout all student journeys



Staff: shared access to catalog, borrows, reservations, AI assistant, notifications, and settings via role-gated middleware.

Frontend Interface — Key Surfaces

Public / Auth

- › /login
- › /verify-email
- › /reset-password
- › /signup
- › /forgot-password

Student

- › /dashboard/catalog
- › /dashboard/reservations
- › /dashboard/history
- › /dashboard/borrowed
- › /dashboard/fines
- › /dashboard/ai-assistant

Instructor

- › /instructor
- › /instructor/reading-lists
- › /instructor/following
- › /instructor/submit-material
- › /instructor/my-submissions

Admin

- › /admin/books
- › /admin/borrows
- › /admin/reports
- › /admin/users
- › /admin/reservations
- › /admin/statistics

Core Library Operations

Reservation Lifecycle



Unique partial index prevents duplicate active reservations per user/book

Scheduler auto-expires reservations and marks overdue borrows

Borrow Lifecycle



Borrow policies configurable per role (max active, days, extensions)

Overdue return → pending fine record created automatically

Fines & Notifications

- Fine states: PENDING · PAID · WAIVED
- Unread notification badge counts
- Types: reservation, borrow, list, system
- Read / delete / clear-read operations

Catalog & Books

- Metadata, ISBN, authors, cover images, tags
- E-book & PDF URLs, max 50 copies/branch
- Faculty, category, subject classification
- External import: OpenLibrary, Gutendex

AI Library Assistant

Powered by OpenRouter · Streaming SSE · Tool Calling · Study Sessions · 5 Response Modes

Model Tiers (Auto-Selected)

FREE	google/gemma-4-31b-it:free	Simple greetings, short Q&A
CHEAP	google/gemini-3.1-flash-lite-preview	Tool calling, catalog queries
SMART	anthropic/claude-3-haiku	Deep reasoning & analysis
STUDY	anthropic/claude-3-haiku (same as SMART)	Study session guide generation

Key Capabilities

- › SSE Streaming chat
- › Auto + manual model select
- › 5 response modes
- › 15 callable tools
- › Book PDF reading
- › Study sessions (per book)
- › Conversation history
- › Admin cover scan (vision)
- › Rate limit: 15 req / 60s

AI Tools & Response Modes

search_catalog

Keyword search across the library catalog

fetch_webpage

Fetch any public URL

get_active_borrows

All active borrows + top 5 most borrowed

get_reading_lists

Published instructor reading lists

list_study_materials

List all indexed study materials

get_book_details

Full details and availability for a book

get_my_borrows

Active borrows and due dates for the user

get_active_reservations

All active reservations (pending/ready)

get_my_reading_lists

Instructor's own lists including drafts

get_chunk_context

Neighbouring chunks around a material chunk

read_ebook

Read an e-book or PDF for summaries/quotes

get_catalog_stats

Total book, copy, and e-book counts

get_user_stats

Registered user counts by role

search_study_material

Full-text search across material chunks

get_material_outline

Opening chunks of a study material

Response Modes:

learning

explanatory

planning

formal

concise

Instructor & Research Materials

Reading Lists

- Create, edit, delete reading lists
- Set visibility (public/private) and status
- Add books with per-item ordering
- Public feed for all authenticated users
- Admin moderation dashboard
- View any instructor's published lists

Follow System

Follow/unfollow instructors · follower counts · following feed

Research Materials

Research Papers

Course Materials

Theses

Professor Publications

Indexing Pipeline:

Upload

Extract Text

Chunk ~400w

Index

FT Search

- Approval & publishing workflow
- Access control: role, faculty, course, public
- Admin reindex & moderation dashboard
- AI tool: search_study_material

Admin Dashboard & Capabilities

Users

- Activate / deactivate accounts
- View all roles and user stats
- User distribution dashboard

Books & Catalog

- CRUD with copy management
- PDF indexing and reindex tools
- AI cover image scan endpoint

Borrows & Reservations

- Full lifecycle management
- Most-borrowed and trend reports
- Scheduler monitoring

Fines & Policies

- Pay or waive fine records
- Per-role borrow policy limits
- Extension count and day rules

Materials

- Approve/reject submissions
- Moderation dashboard
- Reindex pending file queue

Reports & Stats

- PDF and Excel report export
- Admin statistics dashboard
- AI usage metrics endpoint

Functional Requirements

Auth & Access

- Users authenticate before accessing /dashboard/*
- Dashboard routes are role-gated in frontend middleware
- New local users must verify email before login
- Admins can activate and deactivate users

Catalog & Copies

- Books may have multiple physical copies across branches
- Reservations scoped to user / book-copy / branch
- Partial unique index prevents duplicate active reservations
- Maximum 50 copies per branch enforced at API level

Borrow & Fines

- Borrow policies limit active borrows, days, and extensions per role
- Returning an overdue borrow creates a pending fine record
- Scheduler reconciles overdue borrows and expired reservations
- Password reset uses opaque tokens with generic outward messages

AI & Materials

- AI assistant must use tools for all live library data queries
- Assistant must not guess or invent catalog or statistical values
- Materials have publication, approval state, and access-level controls
- API validation strips and rejects non-whitelisted fields

Non-Functional Requirements

Security

- bcrypt password hashing
- HttpOnly JWT cookies — secure + sameSite in production
- Helmet, CORS allowlist, global exception filter
- Rate limiting per endpoint · JWT secret ≥32 chars at startup

Reliability

- Scheduler reconciles overdue borrows and reservations
- S3 config validated at startup — exits if invalid
- Global validation pipe: whitelist + forbidNonWhitelisted
- Health probes: /health/live and /health/ready

Observability

- Request-ID and request-logging on all routes
- Configurable log level and SQL logging
- AI metrics at /dashboard/admin/ai-metrics

Developer Ops

- TypeScript across frontend and backend
- Docker dev targets for both apps
- 14 test suites (unit + e2e)

Not documented: formal uptime/latency targets · RPO/RTO · CI/CD pipeline · accessibility or browser support targets.

Technology Stack & Infrastructure

Frontend

- Next.js 14 App Router
- React 18 + TypeScript
- Tailwind CSS + Framer Motion
- SWR — data fetching
- Radix UI primitives
- Axios + SSE streaming

Backend

- NestJS 10 + TypeScript
- Prisma ORM 5 + PostgreSQL 15
- JWT (HS256) + Passport.js
- Swagger / OpenAPI docs
- @nestjs/throttler rate limits
- bcryptjs · Helmet · CORS

AI & Storage

- OpenRouter (active provider)
- Gemma · Gemini · Claude models
- SSE streaming responses
- PostgreSQL full-text search
- S3 or local file storage
- PDF + DOCX text extraction

Infrastructure

- Docker + Docker Compose
- PostgreSQL 15 + pgAdmin 4
- Node.js 20
- npm workspaces (monorepo)
- pdftkit + exceljs (reports)
- concurrently + wait-on

Testing, Linting & Type Safety

E2E Test Suites

[security.e2e-spec.ts](#)

18 tests — auth guards, rate limits, query validation

[reservations.e2e-spec.ts](#)

20 tests — full reservation lifecycle

[borrows.e2e-spec.ts](#)

8 tests — extend, return, overdue fine

Unit Test Suites (11 files)

- [users.service.spec](#) · [users.controller.spec](#)
- [books.service.spec](#) · [book-document.service.spec](#)
- [reservations.service.spec](#)
- [borrow-scheduler.service.spec](#)
- [ai.service.spec](#) · [catalog-search.service.spec](#)
- [ai-modes.spec](#) · [material-access.util.spec](#) · [global-exception.filter.spec](#)

Verification Commands

`npm run typecheck:api` — TS strict check (API)

`npm run test:api:critical` — 5 critical files, fast

`npm run test:api:e2e` — E2E (needs running DB)

`npm run typecheck:web` — TS strict check (web)

`npm run test:api` — Full unit suite

`cd apps/api && npm run lint` — ESLint + Prettier

Docker, Local Setup & Environment

Docker Compose Services

postgres	:5432	postgres:15-alpine · external volume
api	:3001	NestJS · dev target · waits for postgres
web	:3000	Next.js · dev target · waits for api
pgadmin	:5050	dpage/pgadmin4 · admin@uskudar.edu.tr

Quick Start

1. `docker volume create library-system_edit_postgres_data`
2. `npm install && npm run db:start`
3. `npm run db:migrate && npm run db:seed`
4. `npm run dev → http://localhost:3000`

Critical Environment Variables

JWT_SECRET	≥32 chars · same in API + web
OPENROUTER_API_KEY	Required for all AI features
FRONTEND_URL	e.g. <code>http://localhost:3000</code>
DATABASE_URL	PostgreSQL connection string
CORS_ORIGIN	Allowed frontend origins
STORAGE_PROVIDER	local (default) or s3
SMTP_HOST	Empty → tokens logged to console

Thank You

Questions & Discussion

- 17 slides · Full-stack Node.js monorepo
- NestJS API + Next.js 14 App Router frontend
- OpenRouter AI assistant with 15 callable tools
- 4 user roles · 100+ API endpoints · 14 test suites